CENTRE FOR OCEAN RESEARCH

APPLIED PHYCOLOGY LAB

The Applied Phycology lab with advanced instruments like a Continuous Flow Analyzer (Auto Lab), ATR-FTIR, and Multimode reader explores the potential aspects of algae across different domains including biotechnology, environmental sustainability, and other industrial applications. This lab primarily works on algal adhesion and its consequences in the marine environment to highlight the efforts in developing eco-friendly techniques in combatting marine biofouling problems. Also it focuses on micro algal culture techniques to enhance biomass production and in macro algal population, for spore based seed production.

BIOMOLECULES SEPARATION LAB

The Biomolecules separation lab equipped with Thin-layer chromatography (TLC) and High-performance liquid chromatography (HPLC) used to isolate novel molecules from Natural Resources. This lab provides precise quantitative analysis by separating biomolecules based on their affinity for the stationary phase under high pressure. Through this multi-step process, scientists can effectively separate and analyze biomolecules present in diverse biological samples with accuracy and efficiency. These finds will helpful in designing new compounds or molecules for Environmental and Biomedical application.

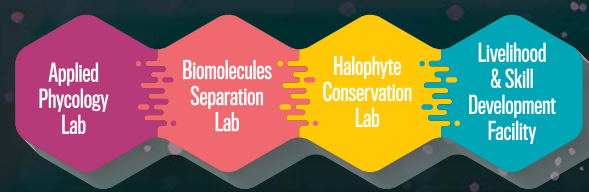
HALOPHYTE CONSERVATION LAB

Halophyte Conservation Lab conducts applied research to perform field and laboratory studies on seagrasses, seaweeds and mangroves. The experiments accomplished will integrate the management strategies for conservation and restoration initiatives of highly productive ecosystems. In-vitro micropropagation and mesocosm chamber in this lab, is used for warming and acidification experiments and to induce halophyte cells by novel methods for the purpose of natural product research, metabolic engineering and conservation aspects.

LIVELIHOOD AND SKILL DEVELOPMENT FACILITY

Livelihood and Skill Development facility is created to empower the youth with adequate skill sets that will enable their employment in relevant sectors and also improve productivity. The facility with associated labs and advanced instrumentation conducts capacity building through Skill development programmes. The trained youth will be encouraged with Institutional Seed fund to develop their own commercial products. In addition, the facility enables to scaleup their finished product to marketable viable product through Sathyabama Technology Business Incubator.

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